

Convex optimization

2 – Linear programming

A special case of convex optimization

2.1 Overview

Definition

- A **linear program** is a special case of a convex optimization problem where:
 - › The objective function is linear
 - › The constraints functions are linear
- One can therefore write a linear program as:

$$\begin{array}{ll} \underset{\mathbf{x} \in \mathbb{R}^n}{\text{minimize}} & f(\mathbf{x}) \\ \text{subject to} & c_i(\mathbf{x}) \leq b_i, \quad i = 1, \dots, m \end{array}$$

where f and $c_1, \dots, c_m$ are linear functions.

About linearity

- The **graph of a linear function** of dimension k is a **hyperplane** of dimension k .
- The set of solutions of a **linear inequality** $c(\mathbf{x}) \leq b$ is thus a **half-space**.
- The opposite of a linear function is also a linear function.
- Consequently: a linear function is both **convex** and **concave**.

Canonical form

- Linear programs can be written in the following form, called **canonical form**:

$$\begin{array}{ll}\text{maximize} & \mathbf{c}^\top \mathbf{x} \\ \text{subject to} & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}\end{array}$$

where $\mathbf{c} \in \mathbb{R}^n$ and $\mathbf{b} \in \mathbb{R}^m$ are vectors of **known coefficients** and $A \in \mathbb{R}^{m \times n}$ is a matrix of **known coefficients**.

An example

- A company seeks to schedule the weeks of engineering invested in two projects A and B in order to maximize profit.
- Project A requires 2 weeks of design, 2 weeks of manufacturing and 3 weeks of quality control for each week of engineering.
- Project B requires 1 week of design, 3 weeks of manufacturing and 1 week of quality control for each week of engineering.
- Considering the current commitments, the design department has 7 weeks available, the manufacturing department has 11 weeks, and the quality control department has 10 weeks.
- The company has calculated that the profit per week of engineering invested is equal to 6 k€ for project A, and 3 k€ for project B.

An example: modeling

$$\begin{array}{ll}\text{maximize} & 6x_1 + 3x_2 \\ (x_1, x_2) \in \mathbb{R}^2 & \\ \text{subject to} & 2x_1 + x_2 \leq 7 \\ & 2x_1 + 3x_2 \leq 11 \\ & 3x_1 + x_2 \leq 10 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$

An example: canonical form

$$\begin{array}{ll}\text{maximize} & \mathbf{c}^\top \mathbf{x} \\ \text{subject to} & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}\end{array}$$

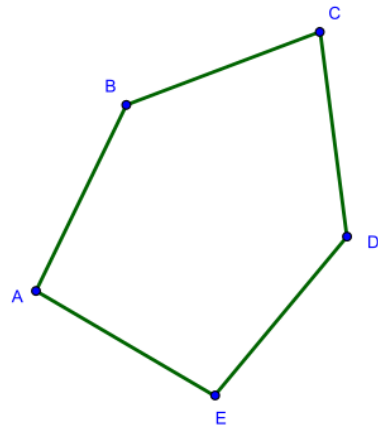
$$\mathbf{c} = \begin{pmatrix} 6 \\ 3 \end{pmatrix} \quad A = \begin{pmatrix} 2 & 1 \\ 2 & 3 \\ 3 & 1 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 7 \\ 11 \\ 10 \end{pmatrix}$$

2.2 Geometric interpretation

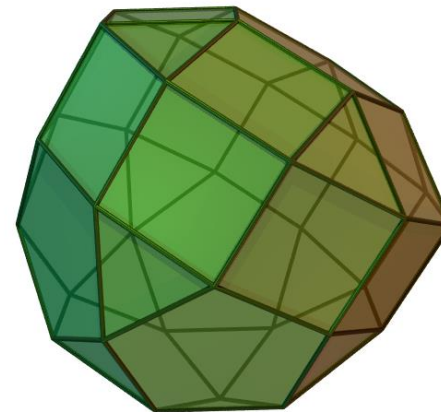
Feasible set of a linear program

- The feasible set of a linear program is defined by several linear inequalities: it is an intersection of **half-spaces**. This set is convex.
- This set describes a **convex polytope** in which the optimal solution is sought.

2-polytope: polygon



3-polytope: polyhedron



Feasible set of our example

maximize
 $(x_1, x_2) \in \mathbb{R}^2$

$$6x_1 + 3x_2$$

subject to

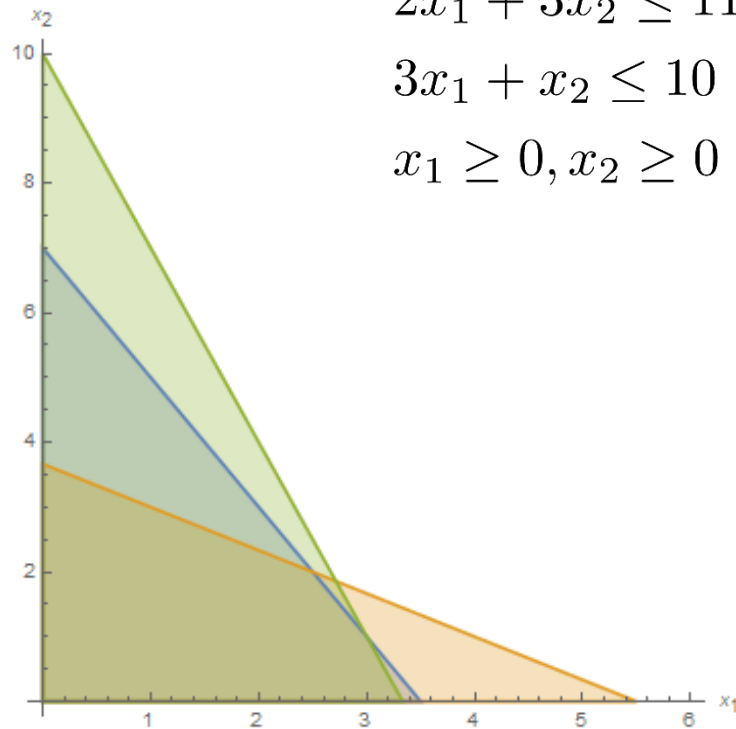
$$2x_1 + x_2 \leq 7$$

$$2x_1 + 3x_2 \leq 11$$

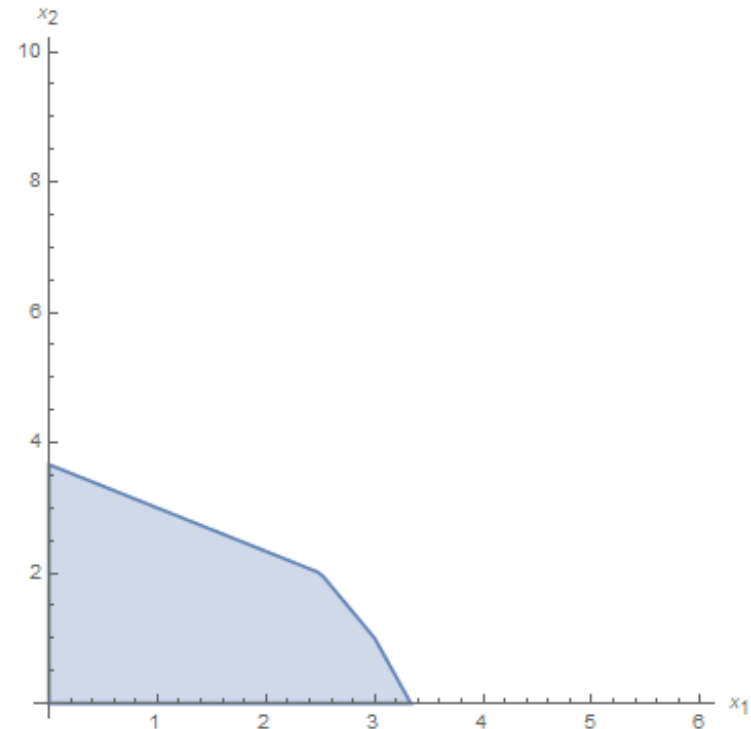
$$3x_1 + x_2 \leq 10$$

$$x_1 \geq 0, x_2 \geq 0$$

Feasible region:



- $2x_1 + x_2 \leq 7$
- $2x_1 + 3x_2 \leq 11$
- $3x_1 + x_2 \leq 10$



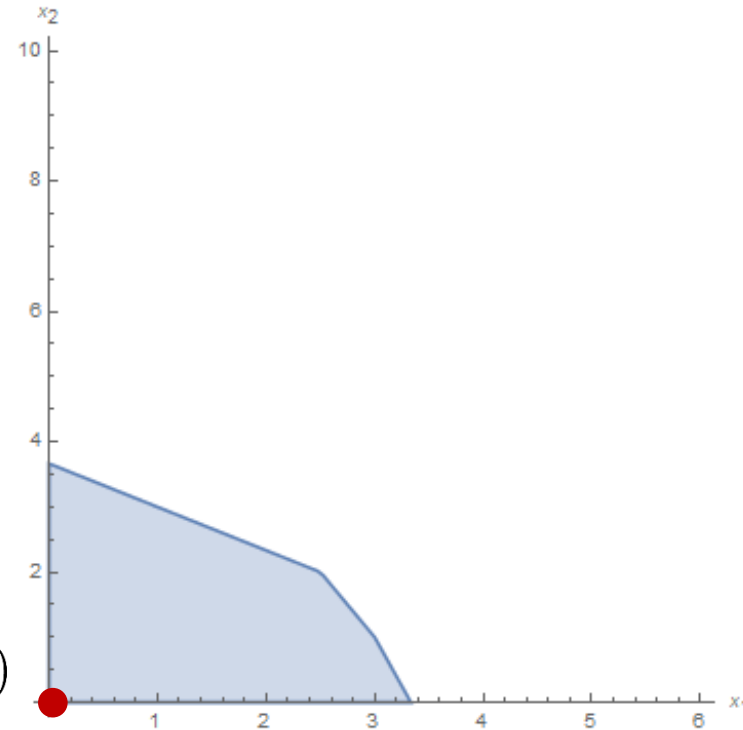
Remarks for resolution

- If the objective function has a maximum value on the feasible region, then it has this optimal value on (at least) **one of the vertices of the convex polytope**.
- The objective function, like the achievable set, is convex. Hence: a local maximum is a global maximum.
- Any idea for a resolution method?

Graphical resolution of our example

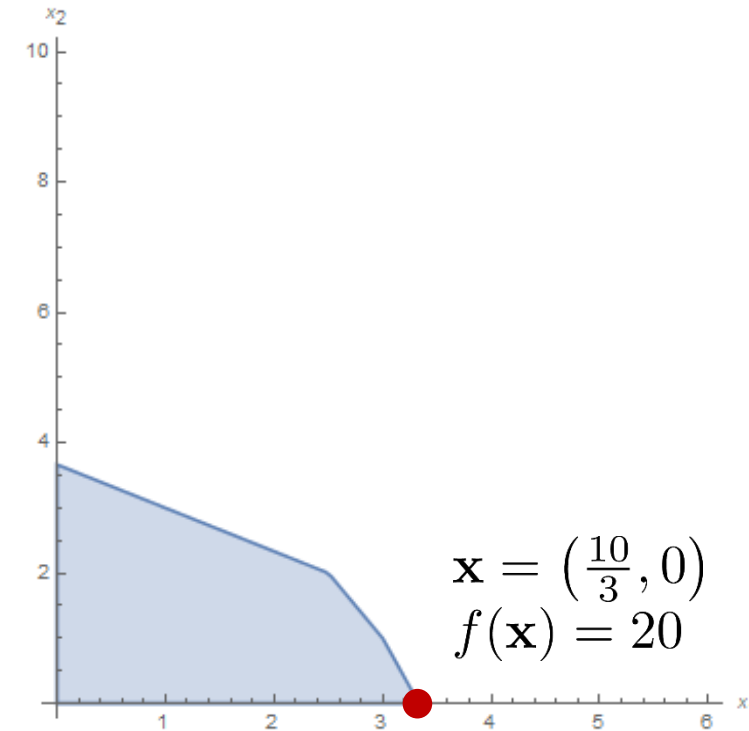
$$\begin{array}{ll}\text{maximize} & 6x_1 + 3x_2 \\ \text{subject to} & (x_1, x_2) \in \mathbb{R}^2 \\ & 2x_1 + x_2 \leq 7 \\ & 2x_1 + 3x_2 \leq 11 \\ & 3x_1 + x_2 \leq 10 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$

$$\begin{array}{l}\mathbf{x} = (0, 0) \\ f(\mathbf{x}) = 0\end{array}$$



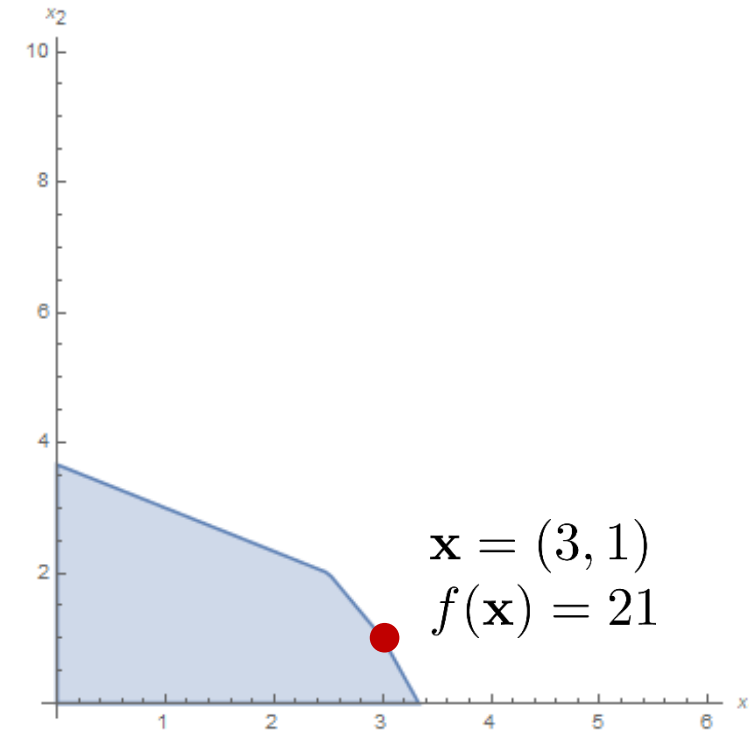
Graphical resolution of our example

$$\begin{array}{ll}\text{maximize} & 6x_1 + 3x_2 \\ (x_1, x_2) \in \mathbb{R}^2 & \\ \text{subject to} & 2x_1 + x_2 \leq 7 \\ & 2x_1 + 3x_2 \leq 11 \\ & 3x_1 + x_2 \leq 10 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$



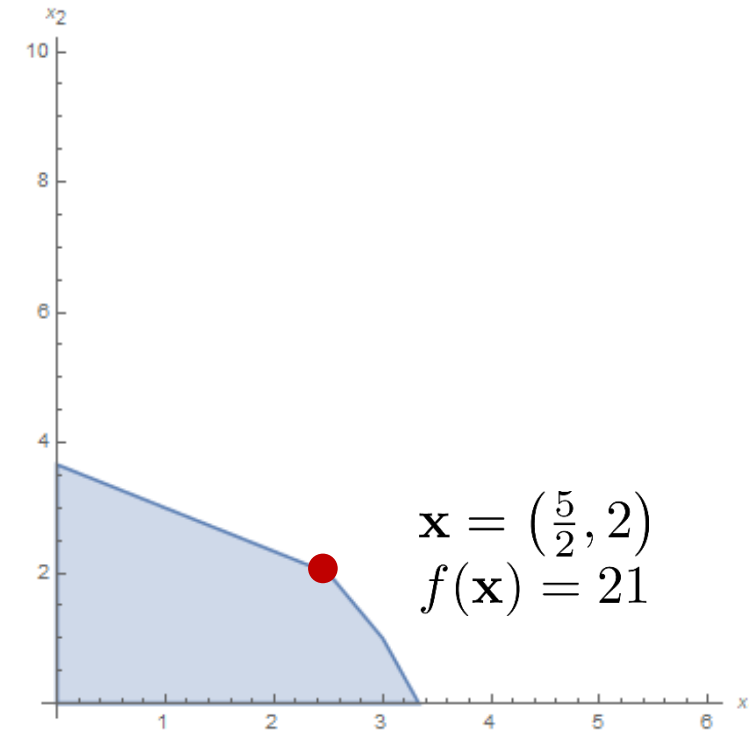
Graphical resolution of our example

$$\begin{array}{ll}\text{maximize} & 6x_1 + 3x_2 \\ (x_1, x_2) \in \mathbb{R}^2 & \\ \text{subject to} & 2x_1 + x_2 \leq 7 \\ & 2x_1 + 3x_2 \leq 11 \\ & 3x_1 + x_2 \leq 10 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$



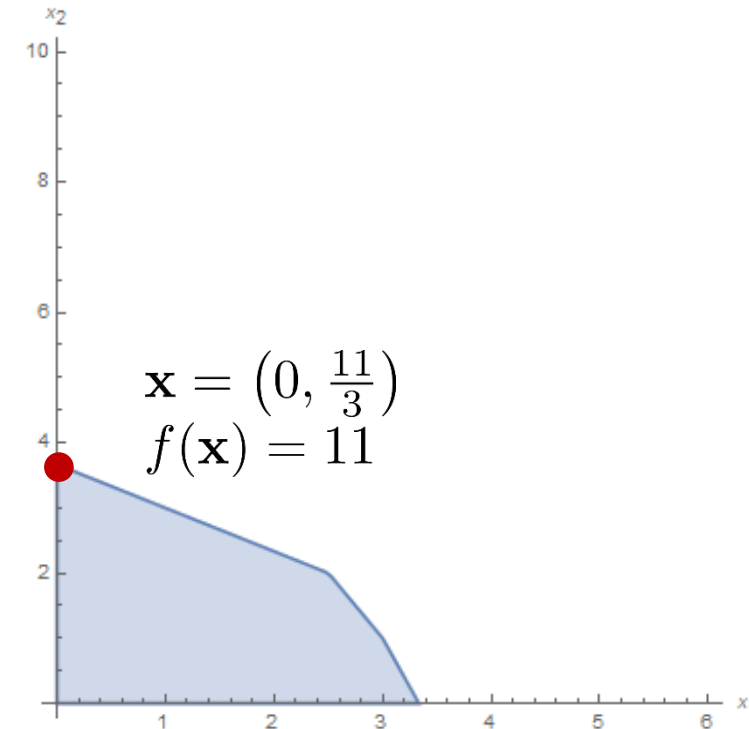
Graphical resolution of our example

$$\begin{array}{ll}\text{maximize} & 6x_1 + 3x_2 \\ (x_1, x_2) \in \mathbb{R}^2 & \\ \text{subject to} & 2x_1 + x_2 \leq 7 \\ & 2x_1 + 3x_2 \leq 11 \\ & 3x_1 + x_2 \leq 10 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$



Graphical resolution of our example

$$\begin{array}{ll}\text{maximize} & 6x_1 + 3x_2 \\ (x_1, x_2) \in \mathbb{R}^2 & \\ \text{subject to} & 2x_1 + x_2 \leq 7 \\ & 2x_1 + 3x_2 \leq 11 \\ & 3x_1 + x_2 \leq 10 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$



Here, the set of optimal solutions is a segment.

2.3 Simplex algorithm

Intuitive description

- The graphical resolution method can be formalized to provide an analytical method: **simplex algorithm**.
- We start from a feasible basic solution.
- We move to **a vertex of the polytope** of the feasible region in the most promising direction.
- Repeat the operation until we reach **a local maximum** (i.e. on a vertex where no direction is interesting to improve the value of the objective function). Convexity implies that this maximum is also **global**. Hence the solution found is **optimal**.

Standard form

- To apply the simplex algorithm, linear problems are written in the following form, called **standard form**:

$$\begin{array}{ll}\text{maximize} & \mathbf{c}^\top \mathbf{x} \\ \text{subject to} & A\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}\end{array}$$

- This form is very similar to the canonical form. It can be obtained from the latter by transforming inequality constraints into equality by adding new variables, called **slack variables**.

Base concept

- To select the most interesting direction, we choose **the variable with the highest coefficient**. The objective of the iteration is then to maximize this coefficient.
- At the same time, the constraints of positivity must be respected. At least one of these constraints **will therefore be limiting** in the choice of the value of this coefficient.
- At each iteration, the problem is rewritten to express the objective function and equality constraints differently, this is the **base concept**.

Base concept

- The in base variables are expressed in function of those out of base (**equality constraints**). Similarly, the objective function is also expressed in terms of variables out of base.
- In an iteration, the chosen variable (the one with the highest coefficient) enters the base. This is the **input base variable**.
- While the variable attached to the limiting constraint for the choice of the value of the input base variable will be removed from the base. This is the **output base variable**.

Summary

1. We write the problem in standard form.
2. We start from a feasible solution (typically with the original variables equal to 0). The slack variables are in base, while the original variables are out of base.
3. The variable with the highest positive coefficient in the objective function is selected to be the input base variable. If all the coefficients are negative then we are at a local maximum: go to 7.
4. The value of the input base variable is maximized while respecting the positivity constraints. The variable attached to the limiting constraint (the one that limits the value) will be the output base variable.
5. The problem (the objective function and constraints) is rewritten with the new base.
6. Loop to 3.
7. We found the global maximum, the variables in base express an optimal solution.

Example

1. We write the problem in standard form.

$$\begin{array}{ll} \text{maximize} & 6x_1 + 3x_2 \\ (x_1, x_2) \in \mathbb{R}^2 & \\ \text{subject to} & 2x_1 + x_2 \leq 7 \\ & 2x_1 + 3x_2 \leq 11 \\ & 3x_1 + x_2 \leq 10 \\ & x_1 \geq 0, x_2 \geq 0 \end{array} \quad \longrightarrow \quad \begin{array}{ll} \text{maximize} & 6x_1 + 3x_2 \\ (x_1, \dots, x_5) \in \mathbb{R}^5 & \\ \text{subject to} & 2x_1 + x_2 + x_3 = 7 \\ & 2x_1 + 3x_2 + x_4 = 11 \\ & 3x_1 + x_2 + x_5 = 10 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0 \end{array}$$

Example

2. We start from a feasible solution (typically with the original variables equal to 0). The slack variables are in base, while the original variables are out of base.

In base: x_3, x_4, x_5

Out of base: x_1, x_2

Feasible solution: $(0, 0, 7, 11, 10)$

$$z = 6x_1 + 3x_2$$

$$x_3 = 7 - 2x_1 - x_2$$

$$x_4 = 11 - 2x_1 - 3x_2$$

$$x_5 = 10 - 3x_1 - x_2$$

Example

3. The variable with the highest positive coefficient in the objective function is selected to be the input base variable.

$$z = 6x_1 + 3x_2$$

$$x_3 = 7 - 2x_1 - x_2$$

$$x_4 = 11 - 2x_1 - 3x_2$$

$$x_5 = 10 - 3x_1 - x_2$$

$\Rightarrow x_1$ is the input base variable

Example

4. The value of the input base variable is maximized while respecting the positivity constraints. The variable attached to the limiting constraint (the one that limits the value) will be the output base variable.

$$\left. \begin{array}{l} (x_3) \quad 7 - 2x_1 \geq 0 \\ (x_4) \quad 11 - 2x_1 \geq 0 \\ (x_5) \quad 10 - 3x_1 \geq 0 \end{array} \right\} \Rightarrow x_1 = \frac{10}{3}$$

$\Rightarrow x_5$ is the output base variable

Example

5. The problem (the objective function and constraints) is rewritten with the new base.

In base: x_3, x_4, x_1

Out of base: x_5, x_2

$$z = 20 + x_2 - 2x_5$$

$$x_3 = \frac{1}{3} - \frac{1}{3}x_2 + \frac{2}{3}x_5$$

$$x_4 = \frac{13}{3} - \frac{7}{3}x_2 + \frac{2}{3}x_5$$

$$x_1 = \frac{10}{3} - \frac{1}{3}x_2 - \frac{1}{3}x_5$$

6. Loop to 3.

Example

3. The variable with the highest positive coefficient in the objective function is selected to be the input base variable.

$$z = 20 + x_2 - 2x_5$$

$$x_3 = \frac{1}{3} - \frac{1}{3}x_2 + \frac{2}{3}x_5$$

$$x_4 = \frac{13}{3} - \frac{7}{3}x_2 + \frac{2}{3}x_5$$

$$x_1 = \frac{10}{3} - \frac{1}{3}x_2 - \frac{1}{3}x_5$$

$\Rightarrow x_2$ is the input base variable

Example

4. The value of the input base variable is maximized while respecting the positivity constraints. The variable attached to the limiting constraint (the one that limits the value) will be the output base variable.

$$\left. \begin{array}{l} (x_3) \quad \frac{1}{3} - \frac{1}{3}x_2 \geq 0 \\ (x_4) \quad \frac{13}{3} - \frac{7}{3}x_2 \geq 0 \\ (x_1) \quad \frac{10}{3} - \frac{1}{3}x_2 \geq 0 \end{array} \right\} \Rightarrow x_2 = 1$$

$\Rightarrow x_3$ is the output base variable

Example

5. The problem (the objective function and constraints) is rewritten with the new base.

In base: x_2, x_4, x_1

Out of base: x_5, x_3

$$z = 21 - 3x_3$$

$$x_2 = 1 - 3x_3 + 2x_5$$

$$x_4 = 2 + 7x_3 - 4x_5$$

$$x_1 = 3 + x_3 - x_5$$

6. Loop to 3.

Example

3. If all the coefficients in the objective function are negative then we are at a local maximum: go to 7.

$$z = 21 - 3x_3$$

$$x_2 = 1 - 3x_3 + 2x_5$$

$$x_4 = 2 + 7x_3 - 4x_5$$

$$x_1 = 3 + x_3 - x_5$$

7. We found the global maximum, the variables in base express an optimal solution $\mathbf{x}^* = (3, 1)$.

Simplex table

- To lighten the notation (and also to facilitate implementation), it is possible to apply the simplex method **using a table** ("**simplex table**").
- In the same way, the in base variables are expressed as a function of the out of base variables.
- The different steps are then applied directly on the table by performing operations on the different lines to rewrite the problem after each iteration with the new base.

Example

1. We write the problem in standard form.

$$\begin{array}{ll} \text{maximize} & 6x_1 + 3x_2 \\ (x_1, x_2) \in \mathbb{R}^2 & \\ \text{subject to} & 2x_1 + x_2 \leq 7 \\ & 2x_1 + 3x_2 \leq 11 \\ & 3x_1 + x_2 \leq 10 \\ & x_1 \geq 0, x_2 \geq 0 \end{array} \quad \longrightarrow \quad \begin{array}{ll} \text{maximize} & 6x_1 + 3x_2 \\ (x_1, \dots, x_5) \in \mathbb{R}^5 & \\ \text{subject to} & 2x_1 + x_2 + x_3 = 7 \\ & 2x_1 + 3x_2 + x_4 = 11 \\ & 3x_1 + x_2 + x_5 = 10 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0 \end{array}$$

Example

2. We start from a feasible solution. The slack variables are in base, while the original variables are out of base.

$$\begin{array}{ll} \text{maximize} & 6x_1 + 3x_2 \\ (x_1, x_2) \in \mathbb{R}^2 & \end{array}$$

$$\text{subject to } 2x_1 + x_2 + x_3 = 7$$

$$2x_1 + 3x_2 + x_4 = 11$$

$$3x_1 + x_2 + x_5 = 10$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0$$



	x_1	x_2	x_3	x_4	x_5	b
x_3	2	1	1	0	0	7
x_4	2	3	0	1	0	11
x_5	3	1	0	0	1	10
z	6	3	0	0	0	0

Example

3. The variable with the highest positive coefficient in the objective function is selected to be the input base variable.

	x_1	x_2	x_3	x_4	x_5	b
x_3	2	1	1	0	0	7
x_4	2	3	0	1	0	11
x_5	3	1	0	0	1	10
z	6	3	0	0	0	0

Example

4. The value of the input base variable is maximized while respecting the positivity constraints. The variable attached to the limiting constraint (the one that limits the value) will be the output base variable.

	x_1	x_2	x_3	x_4	x_5	b	
x_3	2	1	1	0	0	7	$7/2$
x_4	2	3	0	1	0	11	$11/2$
x_5	3	1	0	0	1	10	$10/3$
z	6	3	0	0	0	0	

Example

5. The problem (the objective function and constraints) is rewritten with the new base.

	x_1	x_2	x_3	x_4	x_5	b
x_3						
x_4						
x_1	1	$\frac{1}{3}$	0	0	$\frac{1}{3}$	$\frac{10}{3}$
Z						

$$L_3 \leftarrow \frac{L_3}{3}$$

Example

5. The problem (the objective function and constraints) is rewritten with the new base.

	x_1	x_2	x_3	x_4	x_5	b	
x_3	0	$1/3$	1	0	$-2/3$	$1/3$	$L_1 \leftarrow L_1 - 2L_3$
x_4	0	$7/3$	0	1	$-2/3$	$13/3$	$L_2 \leftarrow L_2 - 2L_3$
x_1	1	$1/3$	0	0	$1/3$	$10/3$	
z	0	1	0	0	-2	-20	$L_4 \leftarrow L_4 - 6L_3$

6. Loop to 3.

Example

3. The variable with the highest positive coefficient in the objective function is selected to be the input base variable.

	x_1	x_2	x_3	x_4	x_5	b
x_3	0	$\frac{1}{3}$	1	0	$-\frac{2}{3}$	$\frac{1}{3}$
x_4	0	$\frac{7}{3}$	0	1	$-\frac{2}{3}$	$\frac{13}{3}$
x_1	1	$\frac{1}{3}$	0	0	$\frac{1}{3}$	$\frac{10}{3}$
z	0	1	0	0	-2	-20

Example

4. The value of the input base variable is maximized while respecting the positivity constraints. The variable attached to the limiting constraint (the one that limits the value) will be the output base variable.

	x_1	x_2	x_3	x_4	x_5	b	
x_3	0	$\frac{1}{3}$	1	0	$-\frac{2}{3}$	$\frac{1}{3}$	1
x_4	0	$\frac{7}{3}$	0	1	$-\frac{2}{3}$	$\frac{13}{3}$	$\frac{13}{7}$
x_1	1	$\frac{1}{3}$	0	0	$\frac{1}{3}$	$\frac{10}{3}$	10
z	0	1	0	0	-2	-20	

Example

5. The problem (the objective function and constraints) is rewritten with the new base.

	x_1	x_2	x_3	x_4	x_5	b
x_2	0	1	3	0	-2	1
x_4						
x_1						
Z						

$$L_1 \leftarrow 3L_1$$

Example

5. The problem (the objective function and constraints) is rewritten with the new base.

	x_1	x_2	x_3	x_4	x_5	b	
x_2	0	1	3	0	-2	1	
x_4	0	0	-7	1	4	2	$L_2 \leftarrow L_2 - \frac{7}{3}L_1$
x_1	1	0	-1	0	1	3	$L_3 \leftarrow L_3 - \frac{1}{3}L_1$
z	0	0	-3	0	0	-21	$L_4 \leftarrow L_4 - L_1$

6. Loop to 3.

Example

3. If all the coefficients in the objective function are negative then we are at a local maximum: go to 7.

	x_1	x_2	x_3	x_4	x_5	b
x_2	0	1	3	0	-2	1
x_4	0	0	-7	1	4	2
x_1	1	0	-1	0	1	3
z	0	0	-3	0	0	-21

7. We found the global maximum, the variables in base express an optimal solution $\mathbf{x}^* = (3, 1)$, with $f(\mathbf{x}^*) = 21$.